Why modulation classifiers fail across transmitters — and a preprocessing fix

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Problem

Deep classifiers for automatic modulation classification reach ~99% on the dataset they were trained on, then lose much of it the moment the transmitter changes. Labelled real RF data being scarce, models are trained on synthetic data and deployed elsewhere — so this gap, not benchmark accuracy, is the practical bottleneck for spectrum monitoring and cognitive radio.

Measurement

A 1-D CNN was trained on RadioML 2018.01A (five modulations shared across domains, 26 SNR levels, ~100k frames of 1024 IQ samples) and evaluated on an independently written signal generator. Gap at SNR $\geq$ 10 dB: **0.182 ± 0.017**, and almost all of it sat in a single cell — 16QAM read as 64QAM 75% of the time.

Diagnosis by controlled elimination — four hypotheses, network frozen so only the test signal varied. Three refuted:

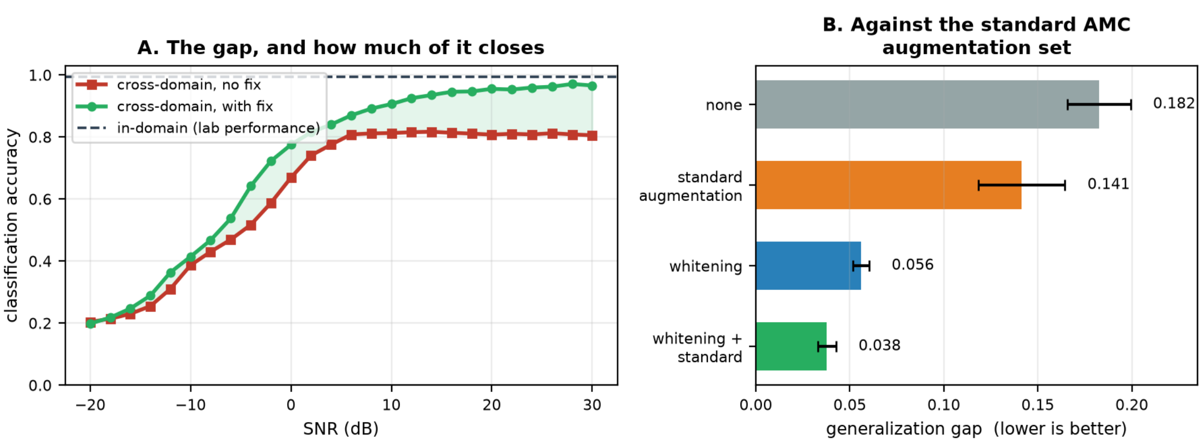
Hypothesis	Test	Outcome
Symbol-rate offset	cyclostationary line driven 14.3 → 0.1 dB	no effect (16QAM 0.168 → 0.202)
Constellation density	amplitude histograms, both domains	refuted — distributions overlap
Channel impairments	CFO, multipath, random phase, combined	refuted — best recovery +0.045
Spectral envelope	phase-preserving magnitude substitution	confirmed — 16QAM 0.191 → 0.972

Fix, and comparison against the literature

Spectral whitening applied as preprocessing to both domains: divide each frame's spectrum by a smoothed estimate of its own magnitude envelope, leaving phase untouched. Unlike adversarial domain adaptation, this needs no target-domain data at fit time, so it is deployable on a receiver that has never met the transmitter it will see. Four seeds, accuracy at SNR $\geq$ 10 dB:

Method	in-domain	cross-domain	gap
none	0.993	0.811	0.182
standard AMC augmentation (rotation / flip / noise)	0.996	0.854	0.141
spectral whitening	0.965	0.909	0.056
whitening + standard augmentation	0.985	0.948	0.038

Whitening beats the standard augmentation baseline by **3.3 standard deviations** and composes with it rather than substituting for it (+6.4 s.d. over whitening alone). **79% of the gap closes.**



A. *Cross-domain accuracy with and without the fix (dashed = in-domain).* ***B.*** *Gap by method, four seeds, bars are one s.d.*

Mechanism — and a hypothesis I had to discard

The obvious account was shortcut learning: the network reads modulation order off the spectral envelope. An envelope-transplant test refutes it — giving frames another class's magnitude spectrum while keeping their own phase leaves the prediction unchanged, with only 0.002 of cases following the donated envelope. What happens instead is out-of-distribution collapse: mismatched frames fall off the training manifold and the classifier defaults to one class, consistently the densest constellation. The errors are systematically biased rather than diffuse — which matters for any monitoring system expected to be trusted in unfamiliar conditions.

Limitations and next step

Both domains are synthetic; there is no real capture yet, and closing that is the main reason I am seeking lab access. Five classes, one architecture, 15 epochs, no comparison against adversarial domain adaptation. The planned next step is hardware: a small programmable RF front end (digital step attenuator, gain block driven into compression, switchable filter bank) that reproduces the same domain variation in measured physical units.